

Algorithms for Big Data

Course Metadata

Credit Structure	L-T-P: 3-0-0 Total: 3
Offering Semester	6 th Sem, Spring
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	NIL
Faculty Names	Jiaul H Paik, Plaban Kumar Bhowmick, Koustav Rudra
Overlapping Subjects	CS60021 (20%)
Justification of overlap	It will be a core subject for BTech in AI program

Course Description

With the growing amount of massive datasets, classical algorithms for processing such datasets often are not very useful. In this course we will focus on algorithmic techniques, and models for big data. The course will introduce common techniques for big data such as randomized methods, sketching, sampling and parallelization on matrices, graphs, static and streaming data.

Course Content

- Introduction to main techniques for large data processing
- Scalable algorithms on large matrices
- Algorithms for large static and streaming data
- Efficient approximate similarity computation
- Compressed sensing and sub-linear time algorithms for testing properties of distributions

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hr (36)
1	Introduction	Introduction, major techniques for big data: randomization, sketching and parallelization	2
2	Algorithms for Matrices	Matrix Chernoff, Subsampled Randomized Hadamard Transform, Approximate Matrix Product, CountSketch, Subspace & Affine Embeddings	6
		Low Rank Approximation, Sketching-based Preconditioning, Leverage Score Sampling, Distributed Low Rank Approximation	5
3	Algorithms for Large Static Graphs	Algorithms Randomized algorithms for mincut and min k-cut, fast triangle counting	3

4	Algorithms for Streams	Estimating Norms in a Data Stream, Heavy Hitters, Graph Streams: Connectivity, Cut, Matchings, Sketching for graphs, k-center clustering in streaming setting	5
5	Similarity Computation	Similarity estimation, Locality sensitive hashing, Approximate Nearest Neighbour Search	4
6	Large Scale Optimization	Smooth convex optimization, distributed SGD, decoupled parallel backpropagation	4
7	Compressed Sensing	Compressed sensing, restricted isometry	3
8	Sublinear time algorithms for Property Testing	Testing Properties of distributions – uniformity, identity, Johnson-Lindenstrass lemma, median estimator	4

Books

1. **Data Streams: Algorithms and Applications**, Muthukrishnan, NOW, 2005
2. **Mining of Massive Datasets**, Leskovec et al, 3e, Cambridge Univ Press, 2020
3. **Randomized Algorithms**, Motwani and Raghavan, Cambridge Univ Press, 1995

References

1. **Multi-Level Local SGD: Distributed SGD for Heterogeneous Hierarchical Networks**, Timothy Castiglia, Anirban Das, Stacy Patterson, ICLR 2021.
2. **Compressive Sensing**, In Handbook of Mathematical Methods in Imaging, Springer, 2011.